EDS 241-Final Project

Case Study: Unilateral climate policies can substantially reduce national carbon pollution (2020)

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Lépissier & Mildenberger (2021)

This analysis aimed to reproduce the results from Lépissier, Alice & Mildenberger, Matto. (2021). The authors employed a synthetic control method (SCM) to estimate the causal effect of the UK's 2001 Climate Change Programme (CCP) on the country's carbon emissions. Their main finding demonstrates that CCP reduced the UK's carbon emissions. CO2 emissions per capita were 9.8% lower in 2005 relative to what they would have been if the CCP had not been passed, demonstrating that unilateral climate policies can still reduce carbon emissions even in the absence of a binding global climate agreement.

The Fundamental Problem & the SMC Fix

The 2001 UK CCP is a complex reform that included a carbon tax on large-scale energy users, industry-negotiated exemptions from the tax for meeting reduction targets, and a voluntary emissions trading scheme. It was one of the first comprehensive climate reform packages passed globally, in advance of action by most other Organisation for Economic Co-operation and Development (OECD) countries.

We are interested in understanding what the UK's carbon emissions might have looked like had the CCP not been enacted.

To do so we need to create a strong counter factual to simulate what *would* have happened. The authors take the synthetic control method (SCM) to manufacture a synthetic UK. This approach is less restrictive than a difference-in-difference strategy would be to compare other countries without a carbon policy.

The synthetic UK is the combination of weighted averages of counties that did not receive the policy intervention. To create the *perfect* synthetic UK, donor country emissions are weighted to match UK pre-treatment emissions trends.

Assessing Treatment Effect

The gap between the synthetic and real UK after policy intervention is observed as the treatment effect.

Before the 2001 policy: Real UK $\approx$ Synthetic UK.

After the 2001 policy: The gap between Real UK and Synthetic UK = *Treatment effect*.

Research Question

Did United Kingdom's 2001 Climate Change Programme (CCP) reduce per capita CO_2 emissions?

- Y (outcome): Per capita CO_2 emissions generated from fossil fuels and the manufacture of cement (power sector) (`EN.ATM.CO2E.PC`)
- Treated unit: United Kingdom (`UK`)
- Treatment year: 2001
- Donor pool: 51 countries with either:
  - (1) in OECD or classified by the World Bank as upper middle–income or high-income countries at the time of treatment in 2001
  - (2) Population greater than 250,000,
  - (3) No carbon pricing policy

How do different donor pools impact the SMC results and robustness checks?

- In this study, we investigate and run SCM for 3 different donor pools. This positions the UK's CCP to be evaluated against a wide pool of donor countries by pre-treatment carbon emissions:
  - (1) Countries with similar economic conditions to UK.
  - (2) High income counties with similar economic conditions to UK.
  - (3) High and upper-middle income counties with similar economic conditions to UK.

The Work Bank classifies countries into income categories according to GNI per capita in US dollars. In fiscal year 2001, the World Bank classified high-income (HIC) countries as those with GNI per capita above 9,265 US$ and upper middle–income (UMC) countries as those with GNI per capita in the 2,996 US$ to 9,265 US$ range. "EN.ATM.CO2E.PC" (CO2 emissions per capita in metric tons).

- additionally, only countries with a population greater than 250,000 without a carbon pricing policy in place in 2001 were considered for the donor pool.

Data

Data Name	Includes
`data`	Carbon emissions data from all accessible countries
`data_OECD`	Carbon emissions from countries with similar economic conditions to UK.
`data_OECD_HIC`	Carbon emissions from high income counties with similar economic conditions to UK.

Data Name	Includes
data_OECD_HIC_UMC	Carbon emissions from high and upper-middle income counties with similar economic conditions to UK.

Study Assumptions

- (1) SUTVA (stable unit treatment values assumption). There is no spillover of CCP enactment to other countries in the donor pool
- (2) Variables used to form the weights must have values for donor pool countries that are similar to those of the UK. This minimizes interpolation bias if the procedure interpolates across regions with very different characteristics
- (3) Strong pretreatment fit between synthetic UK and actual UK. The only credible way to assess treatment effect is by comparing against a strong counter factual, which needs to behave essentially the same as the actual UK.

Approach to address assumptions

- (1) Authors selected the donor sample based on countries that had not passed any major climate legislation until the 2005 European Union's Emissions Trading Scheme (EU ETS). The UK was the only country with the CCP in 2001. This attempts to get the best possible causal effect calculation.
- (2) Using per capita emissions to avoid interpolation bias, meaning emissions data are on the same order of magnitude across pool
- (3) The authors rescale CO2 emissions per capita to a 1990 and 2000 baseline to normalize the pre-treatment window

Recreating UK Synthetic Control

Lépissier & Mildenberger generated synthetic controls using an optimization algorithm to select donor countries to match pre-CCP emissions trajectories in the UK and the synthetic control as closely as possible. This algorithm generates the synthetic UK minimizing mean square predication error (MSPE) of the pre-treatment outcome variable.

We used the {tidysynth} package to generate our synthetic control UK, as compared to the authors' optimized algorithm method which involved a manual econometrics approach. To follow the author's approach, we excluded 1980-1990 from our analysis as they found an inconsistent pre-treatment fit during this pre-treatment period.

```
library(tidysynth) # for synthetic control
library(tidyverse)
library(patchwork) # for combining plots
```

```
library(here)
library(kableExtra)
```

We sourced the original emissions datasets from 1. OECD countries, 2. high-income countries (OECD), and 3. high-income (OECD_HIC) and upper-middle-income countries (OECD_HIC_UMC). These datasets were generated by the authors using World Bank data to classify OECD countries, HIC, and UMC. These filter out countries with populations below 250,000 and carbon policies in place in the study period.

```
# Source loading function to grab all .Rdata objects in files
source(here("R", "load_rdata_files.R"))

# Load in any .Rdata files in directory
load_rdata_files()

# Source  function to filter any donor countries without significant weight
source(here("R", "plot_weights_filtered.R"))


# Filter for post 1990 across all datasets
data_OECD <- data_OECD %>%
  filter(year >= 1990)

data_OECD_HIC <- data_OECD_HIC %>%
  filter(year >= 1990)

data_OECD_HIC_UMC <- data_OECD_HIC_UMC %>%
  filter(year >= 1990)
```

First, we attempted to create a synthetic control that averaged CO2 emissions per capita across the pre-treatment time period by country. We then assessed pre-treatment fit to understand our counter factual usability.

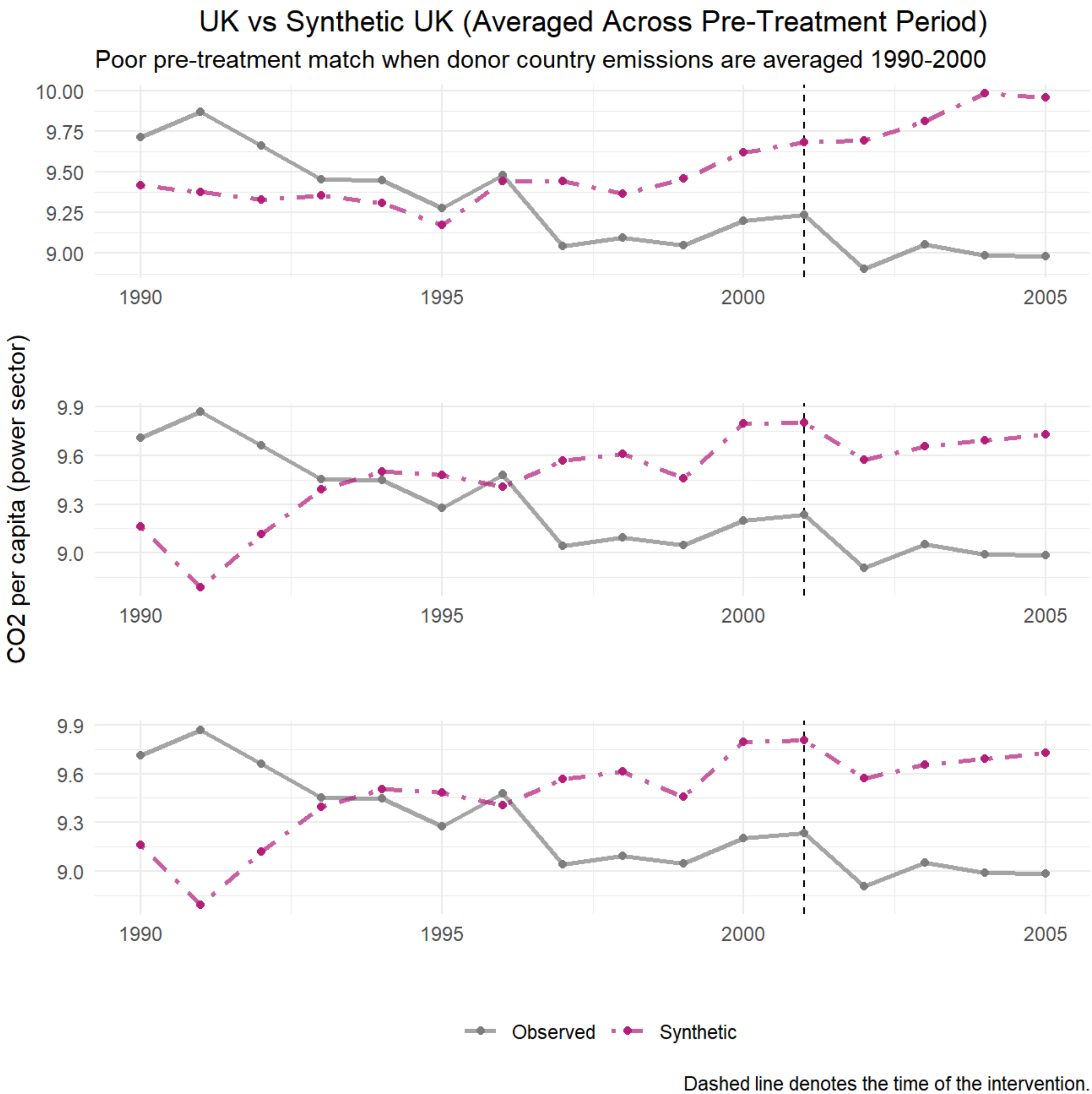
```
# data_OECD
scm_data_OECD0 <- data_OECD %>%
  synthetic_control(
    outcome = `EN.ATM.CO2E.PC`,
    unit = country,
    time = year,
    i_unit = "United Kingdom",
    i_time = 2001,
    generate_placebos = TRUE) %>%

  generate_predictor(  # Predictors (pre-treatment averages)
    time_window = 1990:2001,
    mean.preT = mean(mean.preT, na.rm = TRUE)) %>%

  generate_weights(
    optimization_window = 1990:2001,
    margin_ipop = 0.02,
    sigf_ipop = 7,
```

```
    bound_ipop = 6) %>%
  generate_control() # Generate synthetic control


# data_OECD_HIC
scm_data_OECD_HIC0 <- data_OECD_HIC %>%
  synthetic_control(
    outcome = `EN.ATM.CO2E.PC`,
    unit = country,
    time = year,
    i_unit = "United Kingdom",
    i_time = 2001,
    generate_placebos = TRUE) %>%

  generate_predictor(  # Predictors (pre-treatment averages)
    time_window = 1990:2001,
    mean.preT = mean(mean.preT, na.rm = TRUE)) %>%

  generate_weights(
    optimization_window = 1990:2001,
    margin_ipop = 0.02,
    sigf_ipop = 7,
    bound_ipop = 6) %>%
  generate_control() # Generate synthetic control


# data_OECD_HIC_UMC
scm_data_OECD_HIC_UMC0 <- data_OECD_HIC %>%
  synthetic_control(
    outcome = `EN.ATM.CO2E.PC`,
    unit = country,
    time = year,
    i_unit = "United Kingdom",
    i_time = 2001,
    generate_placebos = TRUE) %>%

  generate_predictor(  # Predictors (pre-treatment averages)
    time_window = 1990:2001,
    mean.preT = mean(mean.preT, na.rm = TRUE)) %>%

  generate_weights(
    optimization_window = 1990:2001,
    margin_ipop = 0.02,
    sigf_ipop = 7,
    bound_ipop = 6) %>%
  generate_control() # Generate synthetic control
```

Pre-Treatment Fit Interpretation In this plot, our synthetic control UK is represented by the pink dashed line to show projected carbon emissions *without the CCP*. The gray line represents the actual observed per capita carbon emissions in the UK post policy implementation.

By plotting the trends of the first resulting synthetic controls across time, we can see that this method results in a poor pre-treatment fit, which violates the assumption of the synthetic control method to make a meaningful causal effect estimation.

```
# data_OECD
p0_data_OECD <- scm_data_OECD0 %>%
  plot_trends() +
  labs(title = "UK vs Synthetic UK (Averaged Across Pre-Treatment Period)",
       subtitle = "Poor pre-treatment match when donor country emissions are averaged 1990-2000",
       y = "",
       x = "",
       caption = NULL)+
  theme(legend.position = "none",
        plot.title = element_text(hjust = 0.5))

# data_OECD_HIC
p0_data_OECD_HIC <- scm_data_OECD_HIC0 %>%
  plot_trends() +
  labs(title = "",
       y = "CO2 per capita (power sector)",
       x = "",
       caption = NULL)+
  theme(legend.position = "none",
        plot.title = element_text(hjust = 0.5))

# data_OECD_HIC_UMC
p0_data_OECD_HIC_UMC <- scm_data_OECD_HIC_UMC0 %>%
  plot_trends() +
  labs(title = "",
       y = "",
       x = "")

# Patchwork plots for each dataset on top of one another
p0_data_OECD / p0_data_OECD_HIC/ p0_data_OECD_HIC_UMC
```



Seconf Synthetic Control Generation

Due to the previous outcome, we generated synthetic outcomes based on yearly emissions by donor country for the most accurate resolution of emissions.

```
run_scm <- function(data) {

  pred_years <- 1990:2000

  pipeline <- data %>%
    synthetic_control(
      outcome     = EN.ATM.CO2E.PC,
```

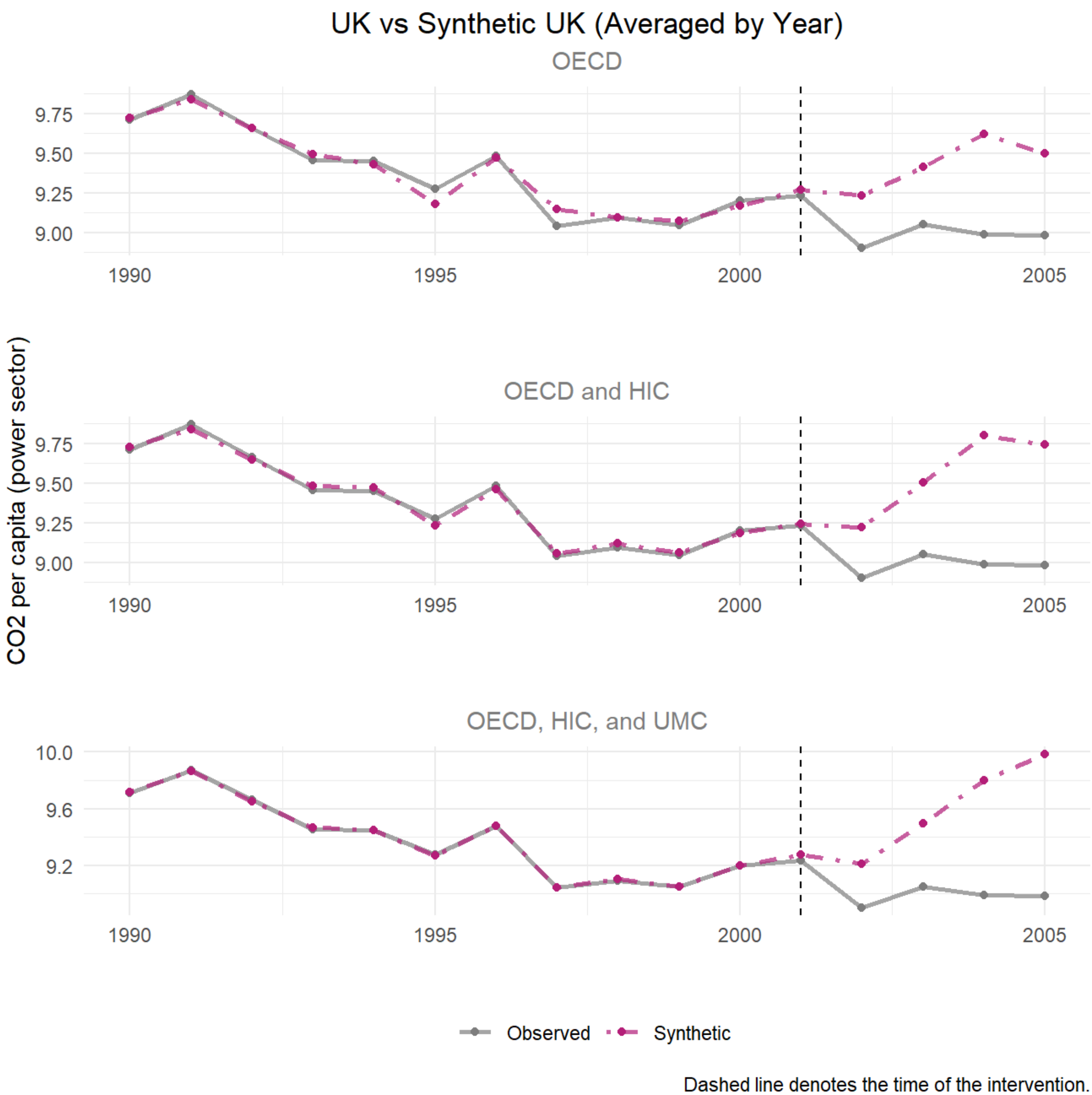
```
      unit       = country,
      time       = year,
      i_unit     = "United Kingdom",
      i_time     = 2001,
      generate_placebos = TRUE)

  for (yr in pred_years) {
    var_name <- paste0("EN.ATM.CO2E.PC_", yr)
    pipeline <- pipeline %>%
      generate_predictor(time_window = yr, !!var_name := mean(EN.ATM.CO2E.PC))
  }

  pipeline %>%
    generate_weights(optimization_window = pred_years) %>%
    generate_control()
}

scm_data_OECD        <- run_scm(data_OECD)
scm_data_OECD_HIC    <- run_scm(data_OECD_HIC)
scm_data_OECD_HIC_UMC <- run_scm(data_OECD_HIC_UMC)
```

Assessing pre-treatment fit

Plotting the outcomes of synthetic control across all 3 donor pool options, we see a much better pre-treatment fit than the previous method of synthetic control. Particularly, our widest donor pool option `OECD_HIC_UMC` presents the strongest pre-treatment fit.

```
# data_OECD
p1_data_OECD <- scm_data_OECD %>%
  plot_trends() +
  labs(title = "UK vs Synthetic UK (Averaged by Year)",
       subtitle = "OECD",
       y = "",
       x = "",
       caption = NULL)+
  theme(legend.position = "none",
        plot.title = element_text(hjust = 0.5),
        plot.subtitle = element_text(color = "grey50", hjust = 0.5))

# data_OECD_HIC
p1_data_OECD_HIC <- scm_data_OECD_HIC %>%
  plot_trends() +
  labs(title = "",
       subtitle = "OECD and HIC",
       y = "CO2 per capita (power sector)",
       x = "",
       caption = NULL) +
  theme(legend.position = "none",
        plot.subtitle = element_text(color = "grey50", hjust = 0.5))

# data_OECD_HIC_UMC
```

```
p1_data_OECD_HIC_UMC <- scm_data_OECD_HIC_UMC %>%
  plot_trends() +
  labs(title = "",
       subtitle = "OECD, HIC, and UMC",
       y = "",
       x = "")+
  theme(plot.subtitle = element_text(color = "grey50", hjust = 0.5))

p1_data_OECD / p1_data_OECD_HIC / p1_data_OECD_HIC_UMC
```



Since the pre-treatment fit assumption is met in our synthetic control recreation, we go on to compare gaps between synthetic UK and observed UK emissions to evaluate treatment effect.

Plot Gaps (Treatment Effect)

Purpose and Meaning of Plotting the Differences

The plotting the gap overlays the outcomes of the real UK against the synthetic UK overtime which make it easy to understand the pre-treatment fit and treatment effect.

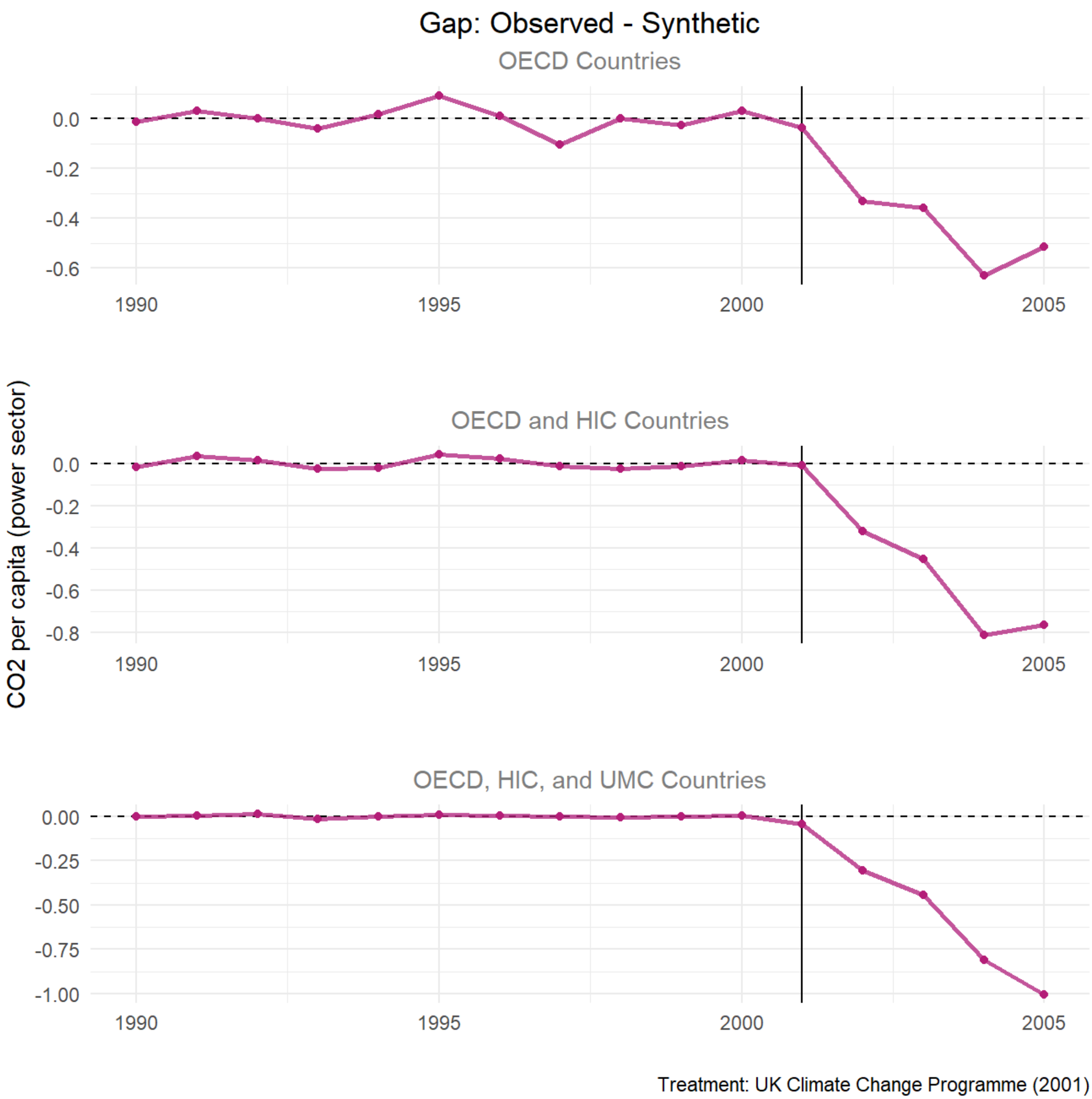
The pre-treatment trends should hover around 0 CO2 emissions per capita displaying a good pre-treatment fit. This means the synthetic control closely tracks the UK's real trends before 2001. The dramatic change in the post treatment shows the divergence of synthetic and real UK outcomes after the policy which is noted as the treatment effect.

Discussion

When looking at the pre-treatment trends, the donor pool with just countries in the OECD has the largest range of pre-treatment trends having a the weakest pre-treatment fit. The donor pool with OECD members, HIC, and UMC shows the best pre-treatment fit as the trend stays close to 0. All three donor pools show a consistently negative and widening gap, meaning UK emissions fall well below synthetic in every specification. Thus, the treatment effect is strong and evident in all scenarios.

```
p2_data_OECD <- scm_data_OECD %>%
  plot_differences() +
  labs(title = "Gap: Observed - Synthetic",
       subtitle = "OECD Countries",
       y = "",
       x = "",
       caption = NULL)+
  theme(legend.position = "none",
        plot.title = element_text(hjust = 0.5),
        plot.subtitle = element_text(color = "grey50", hjust = 0.5))


# data_OECD_HIC
p2_data_OECD_HIC <- scm_data_OECD_HIC %>%
  plot_differences() +
  labs(title = "",
       subtitle = "OECD and HIC Countries",
       y = "CO2 per capita (power sector)",
       x = "",
       caption = NULL)+
  theme(legend.position = "none",
        plot.title = element_text(hjust = 0.5),
        plot.subtitle = element_text(color = "grey50", hjust = 0.5))


# data_OECD_HIC_UMC
p2_data_OECD_HIC_UMC <- scm_data_OECD_HIC_UMC %>%
  plot_differences() +
  labs(title = "",
```

```
       subtitle = "OECD, HIC, and UMC Countries",
       y = "",
       x = "",
       caption = "Treatment: UK Climate Change Programme (2001)") +
  theme(plot.subtitle = element_text(color = "grey50", hjust = 0.5))


# Patchwork plots for each dataset on top of one another
p2_plot <- p2_data_OECD / p2_data_OECD_HIC/ p2_data_OECD_HIC_UMC
p2_plot
```




```
# Save the plot
#ggsave(filename = here::here("figs", "p2_plot_dif.png"),
```

```
      #plot = p2_plot)
```

Placebo Tests

Purpose of the Placebo Tests

The placebo tests assess whether the estimated effect is statistically meaningful by running the SCM for each control unit and plotting their gaps. The real control unit should appear extreme compared to the placebos gaps if the treatment effect is genuine. If there is significance, the control units should not exhibit similar effects as the treated unit.

In this scenario: If the United Kingdom's post-treatment gap is distinctive from the other placebo units, this suggests that the estimated treatment effect reflects a true impact rather than random variation and noise.

Interpreting the plots

- *Pink lines:* The treated unit's (UK) actual gap = the actual UK's CO2 per capita - synthetic UK's CO2 per capita.
- *Grey lines:* The placebo gaps for each control country
- Good *pre-treatment fit* can be noted by all the pre-treatment lines centering around zero.
- *Treatment effect* is displayed by a divergence in the treated unit's post-treatment gap.

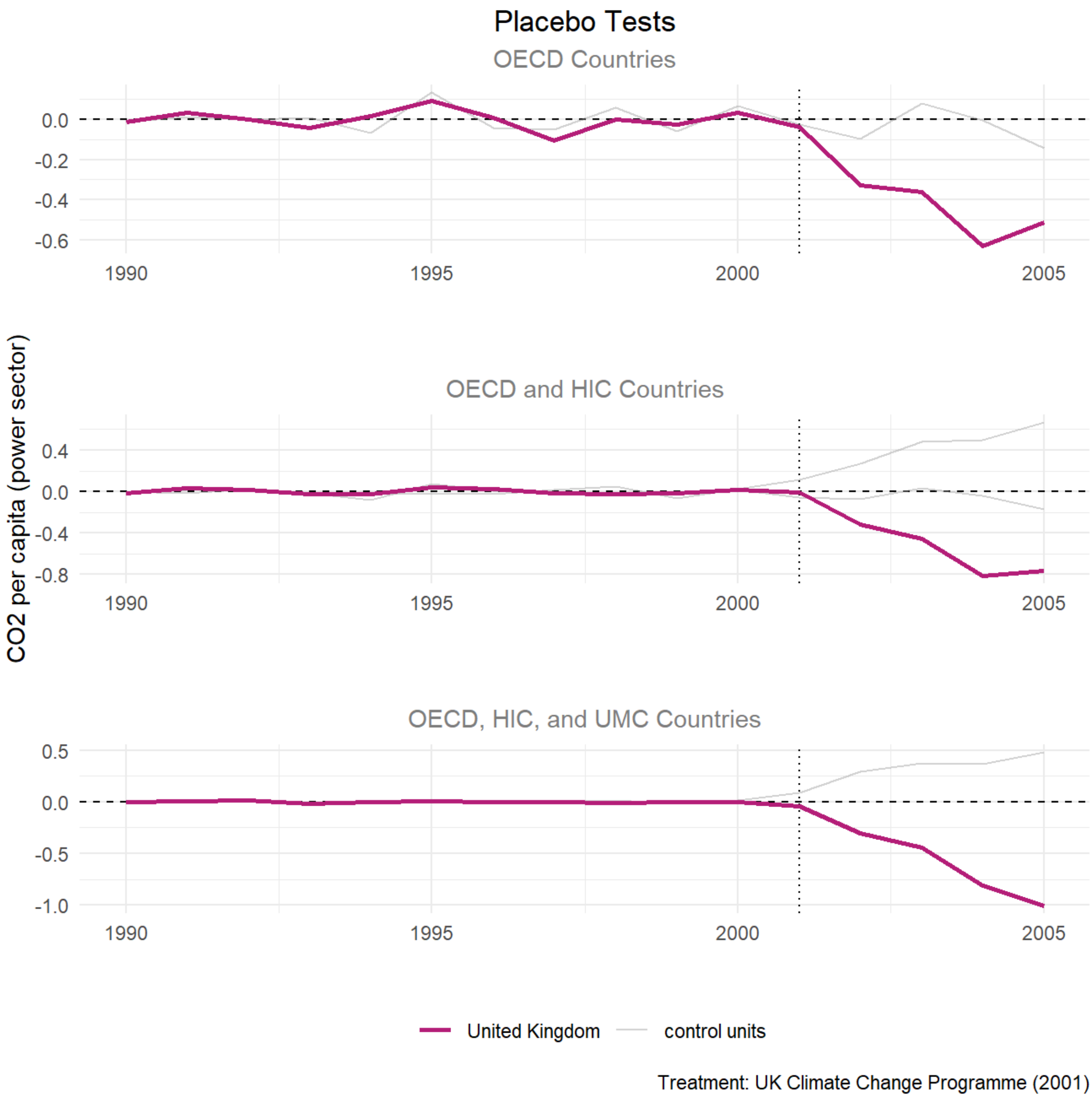
Discussion

All plots show very good pre-treatment fit as the UK and control pre-treatment trends track close to zero. After the intervention, the UK trend dips sharply comparatively to the grey lines, demonstrating a reduction in emissions. The unusual UK's negative gap is evidence for genuine treatment effect, meaning the post-2001 drop in UK emissions is not expected by chance.

The three plots show sensitivity to different donor pool specifications. UK's gap and treatment effect is evident in the placebo test using a donor pool with countries in the OECD (`data_OECD`). However, in the plot where the donor pool has OECD, high income, and upper middle class counties (`data_OECD_HIC_UMC`), UK is partial outlier with some grey lines matching UK's gap. Lastly, in the donor pool with OECD and high income counties (`data_OECD_HIC`), the UK no longer looks like a clear outlier. Here the placebo gaps post-2001 are just as large and/or larger than the UK's. Thus, the treatment effect is less statistically credible under this specification, and most credible in the the OECD-only pool.

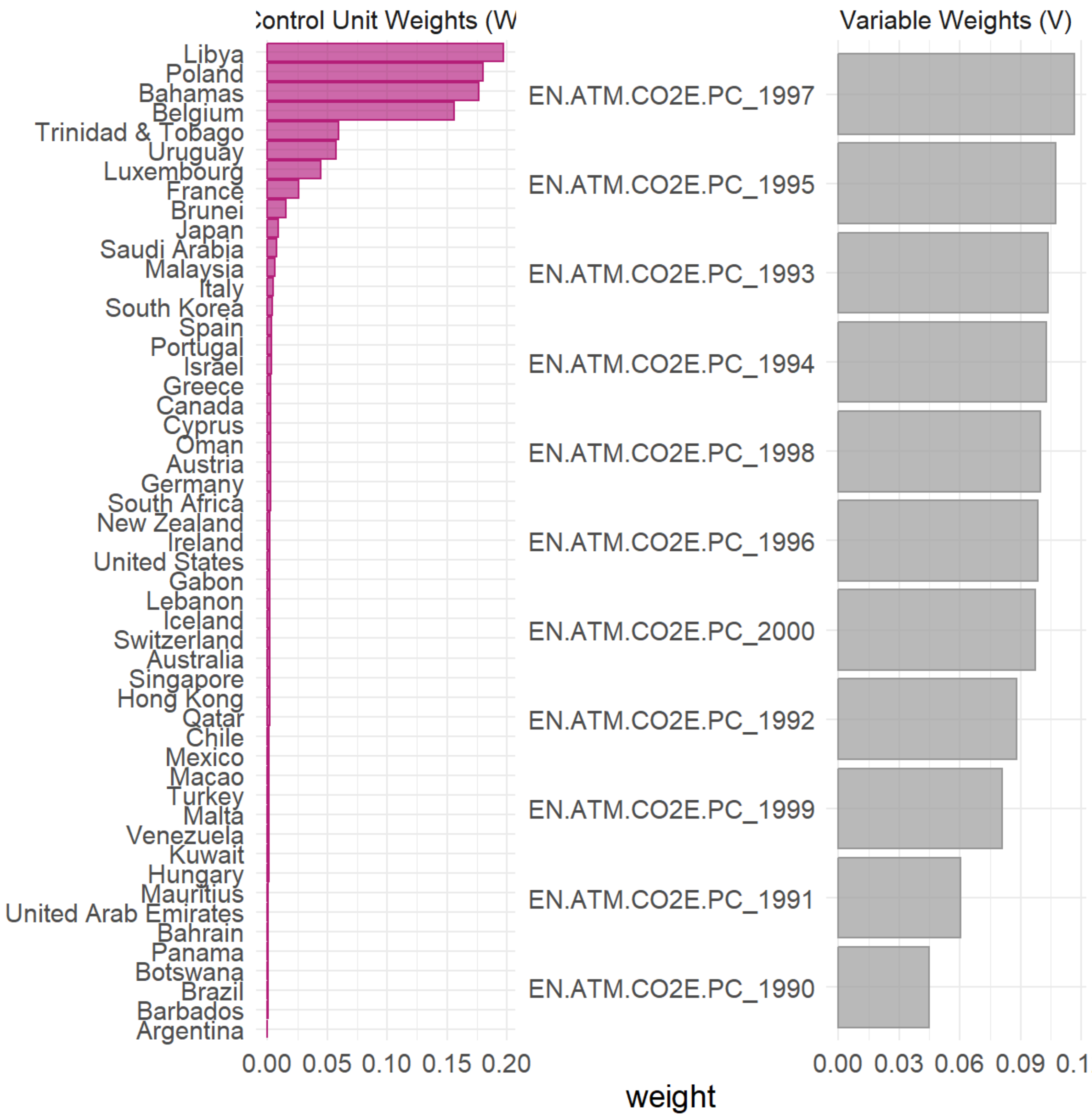
```
p3_data_OECD <- scm_data_OECD %>%
  plot_placebos() +
  labs(title = "Placebo Tests",
       subtitle = "OECD Countries",
       y = "",
       x = "",
       caption = NULL)+
  theme(legend.position = "none",
```

```
         plot.title = element_text(hjust = 0.5),
         plot.subtitle = element_text(color = "grey50", hjust = 0.5))


# data_OECD_HIC
p3_data_OECD_HIC <- scm_data_OECD_HIC %>%
  plot_placebos() +
  labs(title = "",
       subtitle = "OECD and HIC Countries",
       y = "CO2 per capita (power sector)",
       x = "",
       caption = NULL)+
  theme(legend.position = "none",
         plot.title = element_text(hjust = 0.5),
         plot.subtitle = element_text(color = "grey50", hjust = 0.5))


# data_OECD_HIC_UMC
p3_data_OECD_HIC_UMC <- scm_data_OECD_HIC_UMC %>%
  plot_placebos() +
  labs(title = "",
       subtitle = "OECD, HIC, and UMC Countries",
       y = "",
       x = "",
       caption = "Treatment: UK Climate Change Programme (2001)") +
  theme(plot.subtitle = element_text(color = "grey50", hjust = 0.5))


# Patchwork plots for each dataset on top of one another
p3_plot <- p3_data_OECD / p3_data_OECD_HIC/ p3_data_OECD_HIC_UMC
p3_plot
```



Comparing Donor Weights The authors of the original study report their top donor countries as 19% Poland, 19% Libya, 18% Bahamas, 16% Belgium, 6% Trinidad and Tobago, 5% Uruguay, 4% Luxembourg, and 1% Brunei. Additionally, following Based on our gap comparison and placebo tests, our best fit donor pool is the widest OECD, HIC, and UMC pool. Comparing the output from this donor pool's synthetic control, our results are very similar with all the same countries contained, and Libya and Poland representing our top 2 donor countries.

```
plot_weights(scm_data_OECD_HIC_UMC) +
  theme(plot.title = element_text(hjust = 0.5),
    plot.subtitle = element_text(color = "grey50", hjust = 0.5))
```



MSPE Ratio Plot

Purpose and Meaning of MSPE Ratio

The Mean Squared Prediction Error (MSPE)ratio compares post-treatment signal to the pre-treatment error which explains the observed outcome of a unit and its synthetic control (UK and synthetic UK), before and after treatment.

$$\text{MSPE Ratio} = \frac{MSPE_{post}}{MSPE_{pre}}$$

A MSPE ratio of 1 means the intervention did nothing.

A large MSPE ratio means the divergence of treatment gap is unlikely due to chance. Breaking a higher MSPE ratio down further, a smaller denominator means less pre-treatment error, suggesting better synthetic control. And a large numerator means the post-treatment gap is greater than the pre-treatment gap, indicating stronger treatment effect. All to say, a higher MSPE ratio notes that something genuinely changed with the intervention.

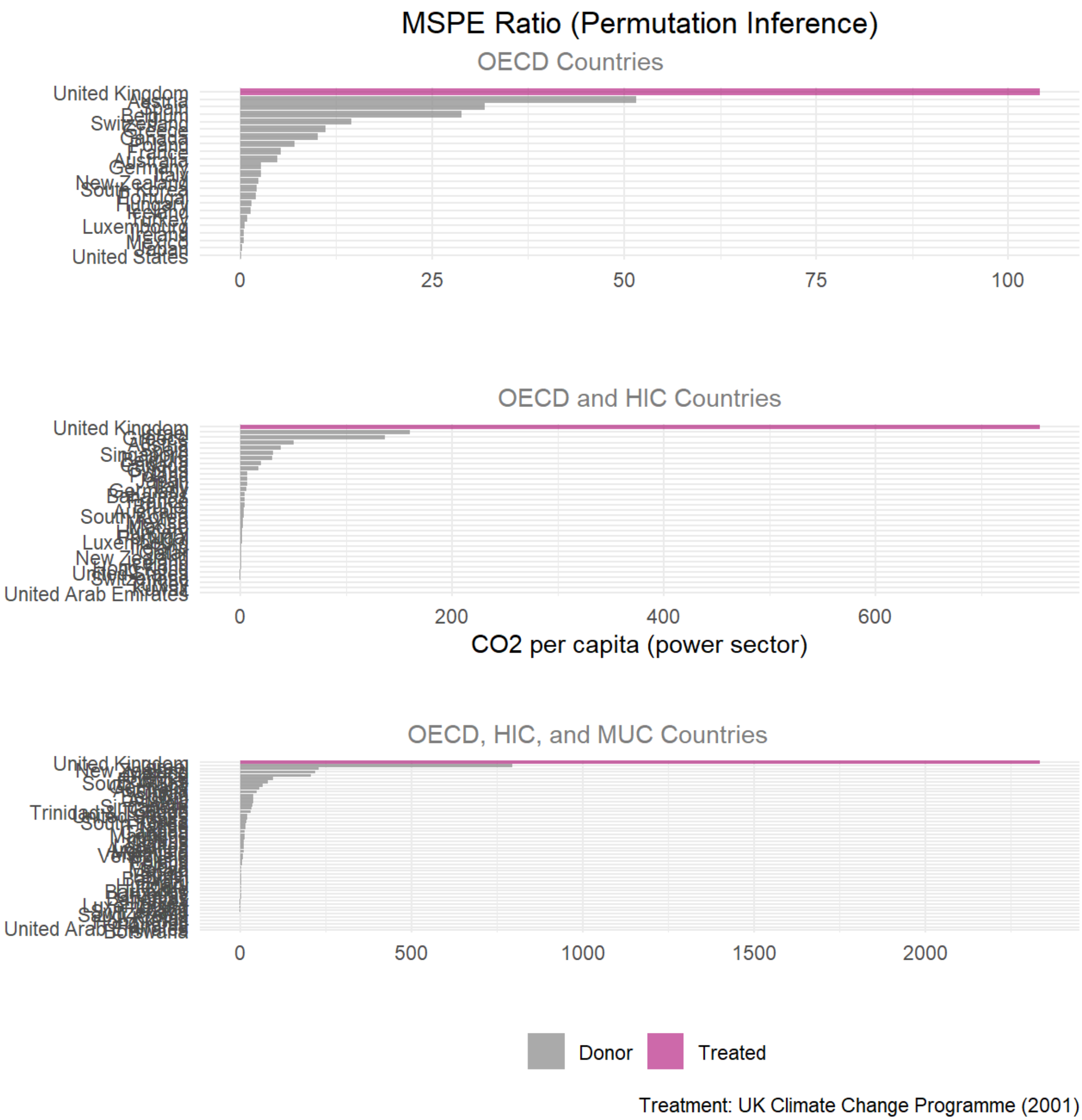
Discussion

When looking at the MSPE plot, it is important to see how the treatment group does in comparison to the other counties by dividing the UK's rank by the total number of countries, which is called the permutation p-value.

Each MSPE ratio plot displays the same trend of the UK ranking #1 with a large gap in MSPE ratio from other countries.

```
p4_data_OECD <- scm_data_OECD %>%
  plot_mspe_ratio() +
  labs(title = "MSPE Ratio (Permutation Inference)",
       subtitle = "OECD Countries",
       y = "",
       x = "",
       caption = NULL)+
  theme(legend.position = "none",
        plot.title = element_text(hjust = 0.5),
        plot.subtitle = element_text(color = "grey50", hjust = 0.4))


# data_OECD_HIC
p4_data_OECD_HIC <- scm_data_OECD_HIC %>%
  plot_mspe_ratio() +
  labs(title = "",
       subtitle = "OECD and HIC Countries",
       y = "CO2 per capita (power sector)",
       x = "",
       caption = NULL)+
  theme(legend.position = "none",
        plot.title = element_text(hjust = 0.5),
        plot.subtitle = element_text(color = "grey50", hjust = 0.5))


# data_OECD_HIC_UMC
p4_data_OECD_HIC_UMC <- scm_data_OECD_HIC_UMC %>%
  plot_mspe_ratio() +
  labs(title = "",
       subtitle = "OECD, HIC, and MUC Countries",
       y = "",
       x = "",
       caption = "Treatment: UK Climate Change Programme (2001)") +
```

```
  theme(plot.subtitle = element_text(color = "grey50", hjust = 0.4))


# Patchwork plots for each dataset on top of one another
p4_plot <- p4_data_OECD / p4_data_OECD_HIC/ p4_data_OECD_HIC_UMC
p4_plot
```




```
OECD_rank <- scm_data_OECD %>%
  grab_significance() %>%
  filter(unit_name == "United Kingdom")

OECD_HIC_rank<- scm_data_OECD_HIC %>%
  grab_significance() %>%
```

```
  filter(unit_name == "United Kingdom")

OECD_HIC_UMC_rank <- scm_data_OECD_HIC_UMC %>%
  grab_significance() %>%
  filter(unit_name == "United Kingdom")


bind_rows(
  grab_significance(scm_data_OECD) %>% filter(type == "Treated") %>% mutate(group = "OECD"),
  grab_significance(scm_data_OECD_HIC) %>% filter(type == "Treated") %>% mutate(group = "OECD_HIC
  grab_significance(scm_data_OECD_HIC_UMC) %>% filter(type == "Treated") %>% mutate(group = "OECD_
) %>%
  select(group, pre_mspe, post_mspe, mspe_ratio, fishers_exact_pvalue, z_score) %>%
  kable(digits = 4, col.names = c("Donor Group", "Pre-MSPE", "Post-MSPE", "MSPE Ratio", "Fisher's
  kable_styling(bootstrap_options = c("striped", "condensed"))
```

Donor Group	Pre-MSPE	Post-MSPE	MSPE Ratio	Fisher's p-value	Z-Score
OECD	0.0022	0.2242	104.0939	0.0435	3.8703
OECD_HIC	0.0005	0.3858	755.0062	0.0303	5.3613

Comparing MSPE outputs for the UK alone across all donor groups, the most minimized MSPE is in the widest donor pool, with a pre-treatment MSPE ratio of 0.0002, which is the closest to the study's finding of a 0.00012 pre-treatment MSPE ratio. We see that against an $a = 0.05$, all donor groups have a significant post treatment gap.

Based on which set of countries we incorporate into our MSPE model, per the different datasets, we see different treatment-effect gaps. Utilizing the data with the most countries, which includes OECD, HIC, and UMC designated countries, we see that the model produces the described result.

Balance Table

Ideally, a properly optimized synthetic control model will have balance, meaning equal or near-equal values across the different predictors between the real United Kingdom and the synthetic version. We use a balance table to compare pre-treatment predictor values the synthetic unit, the real unit, and the donor pool.

```
balance_OECD_HIC_UMC <- scm_data_OECD_HIC_UMC %>%
grab_balance_table()

balance_OECD_HIC_UMC
```

```
# A tibble: 11 × 4
   variable            `United Kingdom` `synthetic_United Kingdom` donor_sample
   <chr>                          <dbl>                      <dbl>        <dbl>
```

```
 1 EN.ATM.CO2E.PC_1990          9.71                         9.71          9.10
 2 EN.ATM.CO2E.PC_1991          9.87                         9.87          8.94
 3 EN.ATM.CO2E.PC_1992          9.66                         9.65          9.35
 4 EN.ATM.CO2E.PC_1993          9.45                         9.47          9.91
 5 EN.ATM.CO2E.PC_1994          9.45                         9.45          9.95
 6 EN.ATM.CO2E.PC_1995          9.27                         9.27          9.90
 7 EN.ATM.CO2E.PC_1996          9.48                         9.48          9.82
 8 EN.ATM.CO2E.PC_1997          9.04                         9.04         10.0
 9 EN.ATM.CO2E.PC_1998          9.09                         9.10         10.0
10 EN.ATM.CO2E.PC_1999          9.05                         9.05          9.91
11 EN.ATM.CO2E.PC_2000          9.20                         9.20         10.3
```

```r
balance_OECD_HIC <- scm_data_OECD_HIC %>%
grab_balance_table()

balance_OECD_HIC
```

```
# A tibble: 11 × 4
   variable            `United Kingdom` `synthetic_United Kingdom` donor_sample
   <chr>                          <dbl>                      <dbl>        <dbl>
 1 EN.ATM.CO2E.PC_1990             9.71                       9.73         11.0
 2 EN.ATM.CO2E.PC_1991             9.87                       9.84         10.6
 3 EN.ATM.CO2E.PC_1992             9.66                       9.65         11.3
 4 EN.ATM.CO2E.PC_1993             9.45                       9.48         11.8
 5 EN.ATM.CO2E.PC_1994             9.45                       9.47         12.0
 6 EN.ATM.CO2E.PC_1995             9.27                       9.23         12.0
 7 EN.ATM.CO2E.PC_1996             9.48                       9.46         11.7
 8 EN.ATM.CO2E.PC_1997             9.04                       9.06         12.0
 9 EN.ATM.CO2E.PC_1998             9.09                       9.12         11.9
10 EN.ATM.CO2E.PC_1999             9.05                       9.06         11.6
11 EN.ATM.CO2E.PC_2000             9.20                       9.19         12.2
```

```r
balance_OECD <- scm_data_OECD %>%
grab_balance_table()

balance_OECD
```

```
# A tibble: 11 × 4
   variable            `United Kingdom` `synthetic_United Kingdom` donor_sample
   <chr>                          <dbl>                      <dbl>        <dbl>
 1 EN.ATM.CO2E.PC_1990             9.71                       9.72         9.37
 2 EN.ATM.CO2E.PC_1991             9.87                       9.84         9.33
 3 EN.ATM.CO2E.PC_1992             9.66                       9.66         9.28
 4 EN.ATM.CO2E.PC_1993             9.45                       9.50         9.30
 5 EN.ATM.CO2E.PC_1994             9.45                       9.43         9.26
 6 EN.ATM.CO2E.PC_1995             9.27                       9.18         9.13
 7 EN.ATM.CO2E.PC_1996             9.48                       9.47         9.40
 8 EN.ATM.CO2E.PC_1997             9.04                       9.15         9.40
 9 EN.ATM.CO2E.PC_1998             9.09                       9.09         9.33
```

```
10 EN.ATM.CO2E.PC_1999          9.05                        9.07         9.42
11 EN.ATM.CO2E.PC_2000          9.20                        9.17         9.58
```

Balance Table Discussion

As seen in the balance sheet outputs, there is a strong balance between all three columns across the different predictors. In particular, we see extremely close results between the UK and its synthetic version. This indicates that the synthetic control closely mirrors the real UK across all of the pre-treatment predictors.

Final Plot

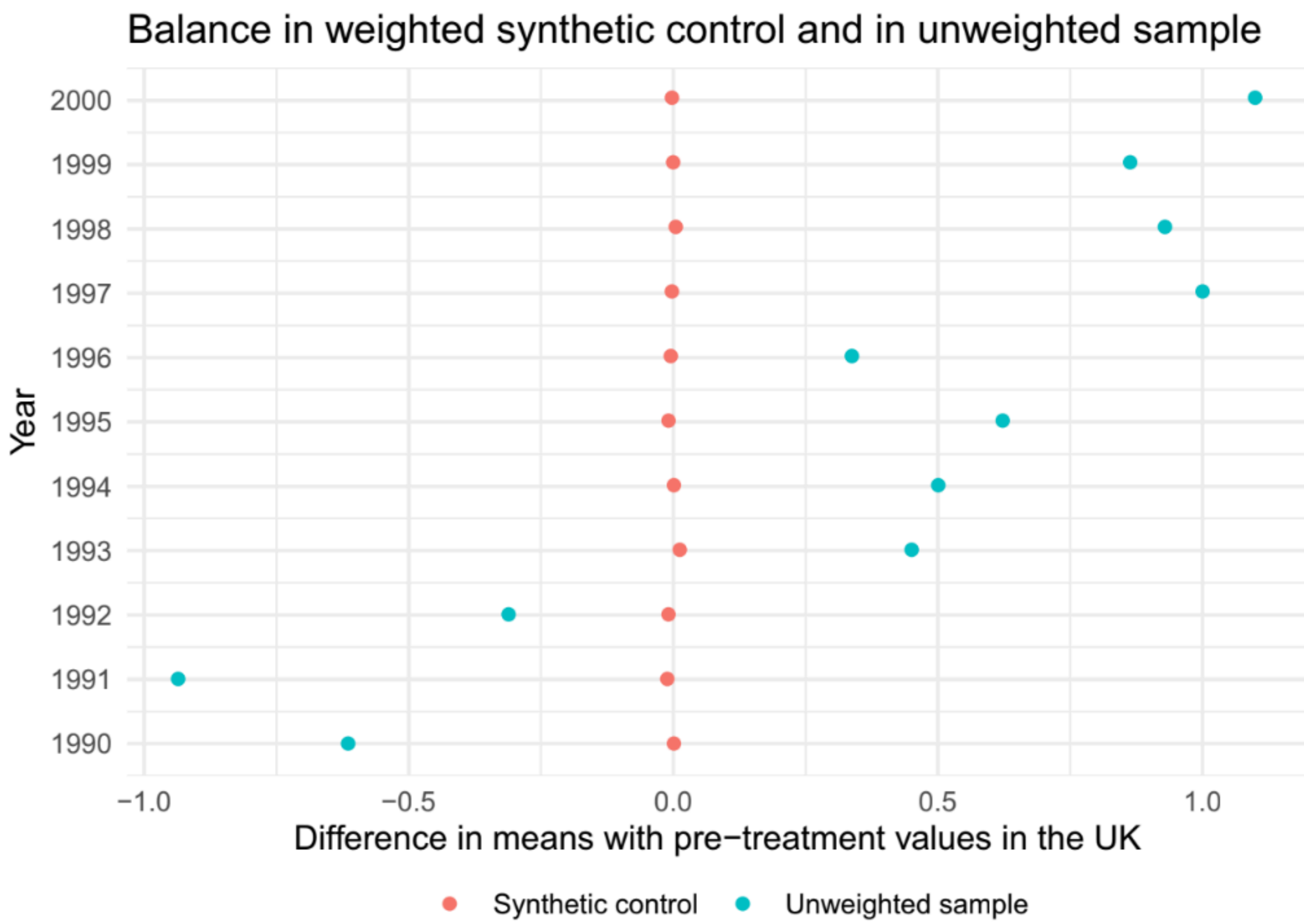
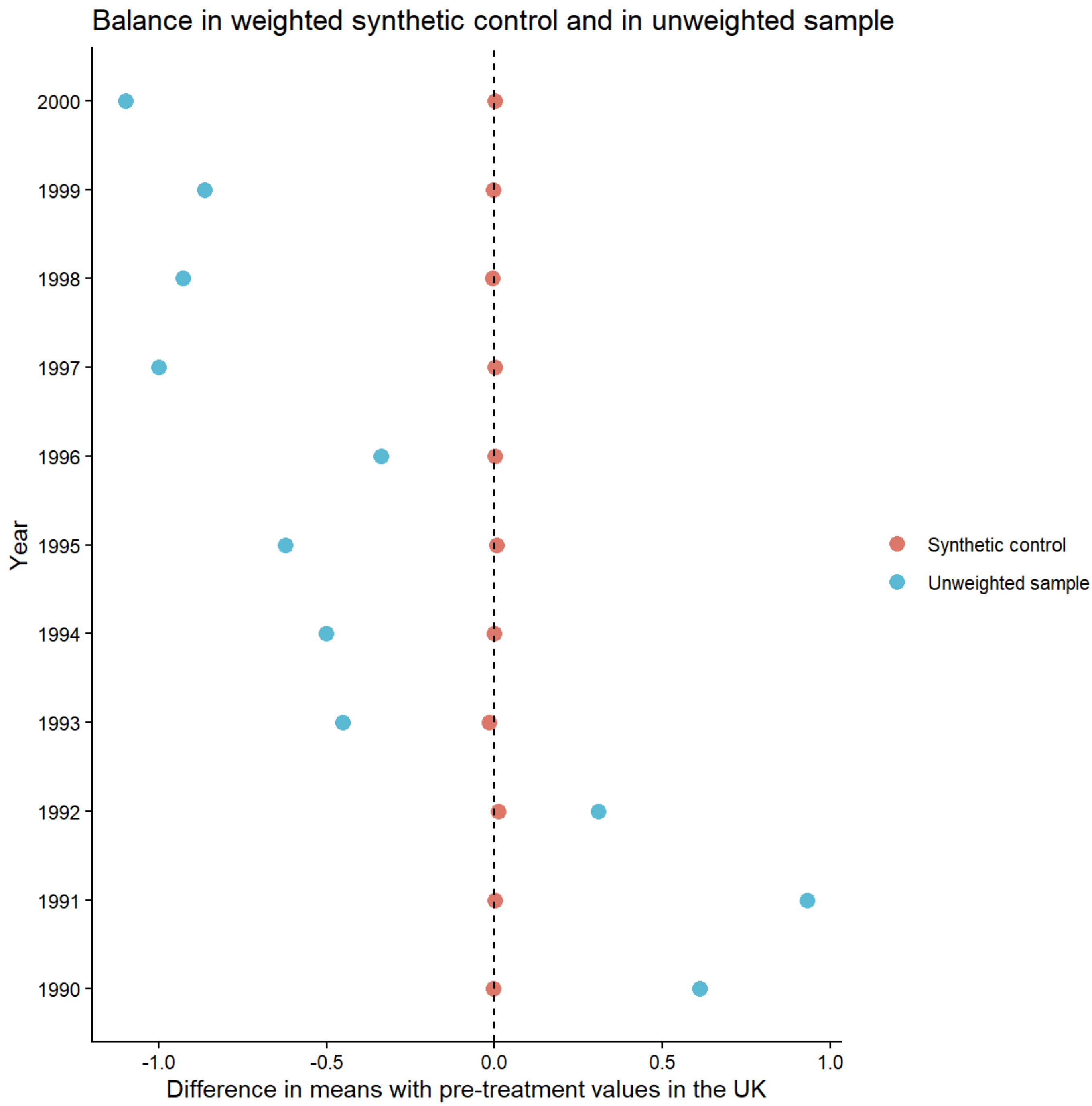


Fig. 2 Difference in means in pre-treatment values observed in the UK and those estimated by the synthetic control (in orange) which is a weighted sample of the donor pool comprised of OECD, high-, and upper middle–income countries. Blue points represent the difference in means in pre-treatment values observed in the UK and those observed in the same donor pool sample, but unweighted

We recreated the difference in pre-treatment CO2 emissions, and displayed the weighted and weighted means in reference to Figure 2 in Lépissier & Mildenberger. This figure shows that our synthetic control model closely matches the pre-treatment balance found in the original paper.

```
scm_data_OECD_HIC_UMC %>%
    # Extract values from balance table
    grab_balance_table() %>%
    # Get differences between actual UK and synthetic and compare to donor differences
```

```
mutate(
    synth_diff = `United Kingdom` - `synthetic_United Kingdom`,
    donor_diff = `United Kingdom` - donor_sample,
    year = as.integer(str_extract(variable, "\\d{4}"))
) %>%
# Make into a tidy data format
pivot_longer(
    cols = c(synth_diff, donor_diff),
    names_to = "type",
    values_to = "diff"
) %>%
# Recode values to match those in paper result
mutate(type = recode(type, "synth_diff" = "Synthetic control", "donor_diff" = "Unweighted sam
# Plot
ggplot(aes(x = diff, y = factor(year), color = type)) +
geom_point(size = 3) +
geom_vline(xintercept = 0, linetype = "dashed") +
# Match colors used by paper figure
scale_color_manual(values = c(
    "Synthetic control" = "#E07B6A",
    "Unweighted sample" = "#5BBCD6"
)) +
labs(
    x = "Difference in means with pre-treatment values in the UK",
    y = "Year",
    # Take out legend title the lazy way
    color = "",
    title = "Balance in weighted synthetic control and in unweighted sample"
) +
# Get all the lines out
theme_classic()
```



Conclusion

We have produced a very near recreation of the original Lépissier and Mildenberger study. We utilized an existing package `tidysynth` to create our synthetic control model. While initially, we had attempted to average the CO2 emissions per capita across the pre-treatment time window, by country. This, however, led to poor model fit, and thus would not serve as an accurate counter factual.

To create a more accurate synthetic control, we amended our model, and generated synthetic outcomes based on yearly emissions by donor country for the most accurate resolution of emissions. This provided much more accurate results that closely followed the real emission levels within the UK. With our more confident synthetic model, we progressed to modeling the effect of the treatment.

Utilizing all three donor pool sets, we modeled the gap, and found that within the study’s target years of 1990-2005, our model produces a very accurate recreation of the true per capita emissions with the UK. From there we then ran a series of placebo tests across the same data sets, and found strong match with these as well.

From there, we plotted the MSPE ratios, which show the effect of the treatment in comparison to the pre-treatment error. While we saw differences in the resulting ratio between the different donor pools, we saw a strong ratio value with the pool used within the study. Specifically, the donor pool including OECD, HIC, and UMC countries. This high ratio, with the donor pool utilized in the study, gives us confidence in stating that the change in carbon emissions we modeled is a causal outcome of the treatment.

As a final robustness check, we created balance tables of the the three different data sets to determine match between the synthetic and real UK. We found extremely close values across all the predictors. While there was some variation in balance between the two UKs and the respective donor pool, depending on which pool of countries was used, there was a consistent and close matching in each table between the synthetic and real United Kingdoms.

To show our model’s results, we recreated the pre-treatment means of the weighted and unweighted per-capita emissions, and found that our results closely match the results shown in Figure 2 of the paper.

In summary, using different means to achieve a SCM method, we have recreated a similar synthetic control model, showed accurate fit between our model and the real UK, and produced results very close to those found within the paper.